

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Both MMLU and the deployment use text output, satisfying basic modality match. However, MMLU's classification-accuracy metric over A/B/C/D tokens [\[Q49, Q56\]](#) cannot evaluate the open-ended, multi-perspective explanatory responses the deployment's tourist/expat assistant produces. Calibration evaluation [\[Q64–Q67\]](#) is a useful auxiliary signal — GPT-3's 24% confidence-accuracy gap [\[Q66\]](#) is informative — but does not address the form mismatch. With OF rated MODERATE, this is a real but partial concern: MMLU can score factual recall in MCQ form but not the explanatory output form the deployment expects.

ID	Evidence
Q64	"We should not trust a model's prediction unless the model is calibrated, meaning that its confidence is a good estimate of the actual probability the prediction is correct." (p.7)
Q65	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q66	"Its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects." (p.7)
Q67	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019)." (p.7)
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with 'The following are multiple choice questions (with answers) about [subject].' For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with 'Answer: '. The model then produces probabilities for the tokens 'A,' 'B,' 'C,' and 'D,' and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)

Strengths

- Text output modality matches deployment text interface
- Calibration metrics (RMS calibration error, confidence-accuracy correlation) [\[Q65, Q67\]](#) provide useful auxiliary signals on whether models 'know what they don't know' [\[Q16\]](#) — directly relevant to a tourist/expat assistant that should hedge on contested topics
- Zero-shot/few-shot evaluation regime [\[Q3\]](#) aligns with deployment's open-domain generalization requirement (Strength 4)

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Q65	"We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject." (p.7)
Q67	"Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019)." (p.7)
Q16	"Worryingly, we also find that GPT-3 does not have an accurate sense of what it does or does not know since its average confidence can be up to 24% off from its actual accuracy." (p.2)
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Modality matches (text), but output granularity mismatches: MCQ token probability [Q56] vs. open-ended explanation expected by deployment users.

OF-2: Not applicable — text-based deployment, no TTS requirement documented.

OF-3: Tourist/expat target population is literate adult readers; literacy not a concern. INSUFFICIENT DOCUMENTATION on accessibility-specific output requirements.

OF-4: Form mismatch documented: MCQ scoring [Q49] cannot credit appropriately-hedged multi-perspective open-ended responses; the authors note their evaluation format 'is not identical to the format in which information is acquired during pretraining' [Q76].

ID	Evidence
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q76	"Importantly, the format of our evaluation is not identical to the format in which information is acquired during pretraining." (p.8)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Whether the deployed system generates free-form text or also offers MCQ-style confirmations (interface design not specified in YAML)

Requires Expert Verification

- Whether deployment evaluation will supplement MMLU MCQ accuracy with open-ended generation metrics (e.g., MENAValues-style prompting)

Remediation

Gap 1: MCQ accuracy metric cannot evaluate open-ended explanatory or perspective-acknowledging responses that the deployed assistant produces.

Recommendation: Add open-ended generation evaluation with rubric-based scoring (cultural appropriateness, multi-perspective acknowledgment, calibrated hedging on contested topics). Re-evaluate every 6–12 months given rapid Arabic-benchmark ecosystem growth [WEB-12].

ID	Evidence
WEB-12	Survey of Arabic benchmarks notes gaps in coverage of North Africa, Levant, Iraq even within the Arabic NLP ecosystem (https://arxiv.org/html/2510.13430v1)

